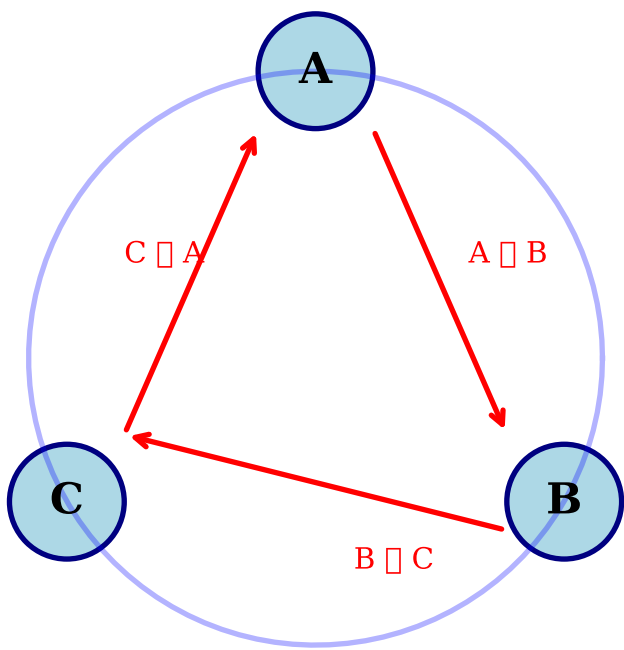
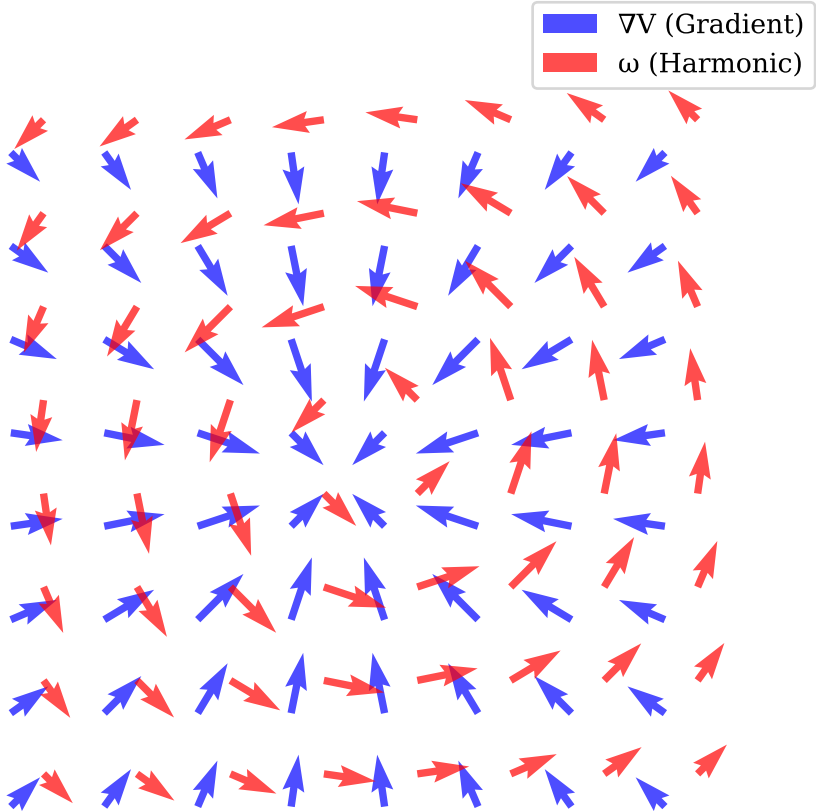


**(A) Condorcet Cycle Detection
via Sheaf Cohomology**



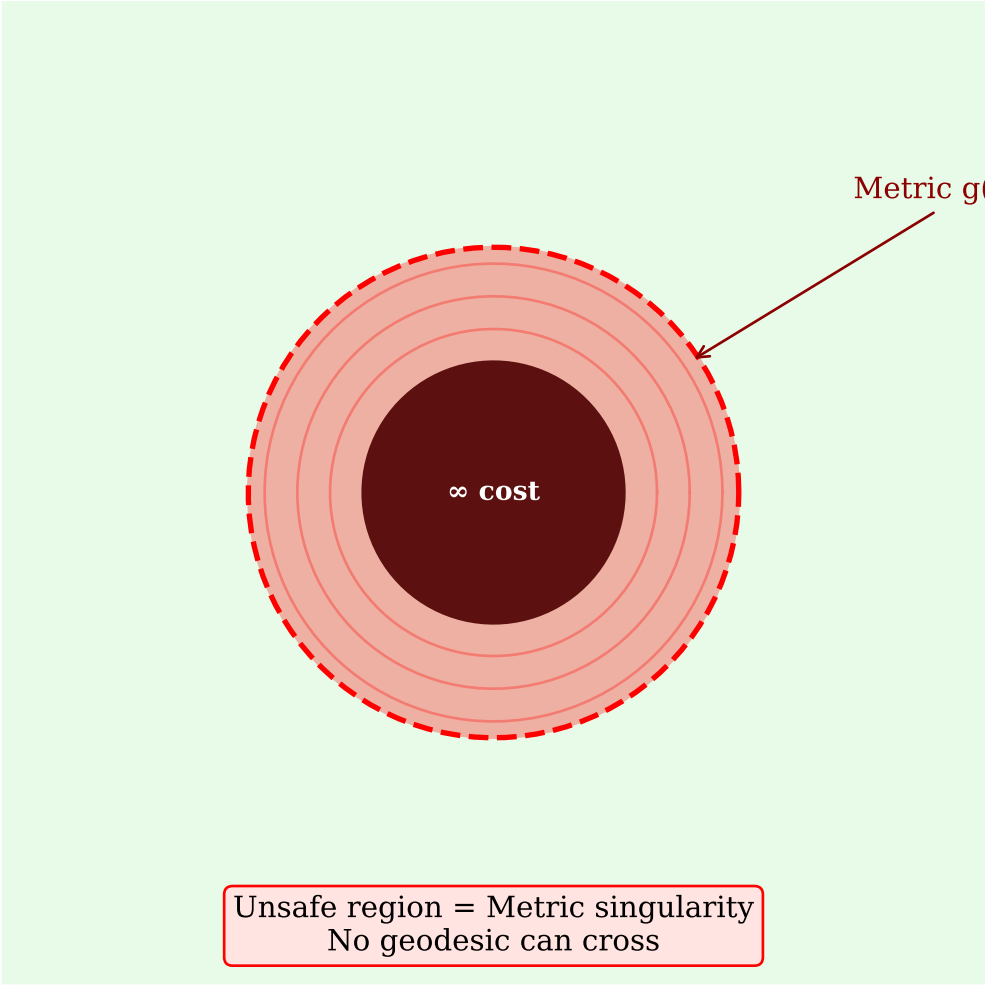
$H^1 \neq 0$: Cyclic preferences detected

**(B) Hodge Decomposition
Separates Learnable from Cyclic**

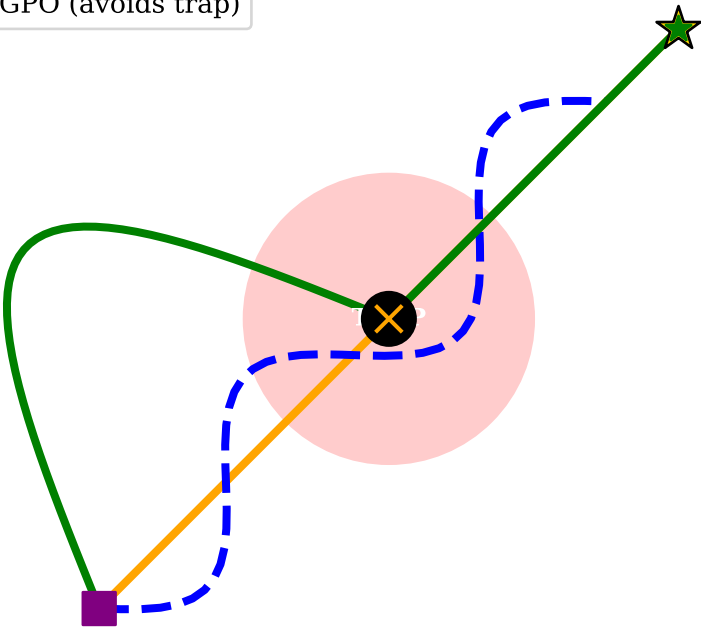
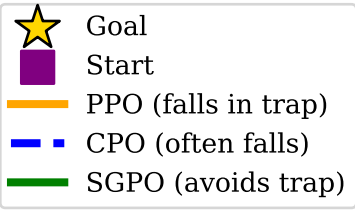


Reward = $\nabla V + \omega$
Learnable + Irreducible Cycle

**(C) Black Hole Safety Constraint
Geometric Barrier (Not Penalty)**



**(D) Policy Trajectories
SGPO Navigates Around Danger**



SGPO: Geodesic path cannot cross singularity
→ 0% safety violations